


```

R2(config-if)#no sh
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int e0/1
R2(config-if)#no ssh
R2(config-if)#no s
R2(config-if)#no sh
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int e0/2
R2(config-if)#ip add 23.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int e0/3
R2(config-if)#ip add 24.0.0.1 255.0.0.0
R2(config-if)#no ssh
R2(config-if)#no sh
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#
R2(config)#
R2(config)#exit
R2#wri
*Aug 26 11:47:06.327: %SYS-5-CONFIG_I: Configured from console by console
R2#write
Building configuration...
[OK]
R2#
*Aug 26 11:48:39.589: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wiz
R2#

```

```

Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 192.50.20.0 255.255.255.0 12.0.0.1
R2(config)#ip route 223.0.50.0 255.255.255.0 26.0.0.2
R2(config)#ip route 32.0.0.0 255.0.0.0 23.0.0.2
R2(config)#ip route 45.0.0.0 255.0.0.0 24.0.0.2
R2(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.2
R2(config)#exit

```

```

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/8 is directly connected, Ethernet0/0
L       12.0.0.2/32 is directly connected, Ethernet0/0
    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       23.0.0.0/8 is directly connected, Ethernet0/2
L       23.0.0.1/32 is directly connected, Ethernet0/2
    24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       24.0.0.0/8 is directly connected, Ethernet0/3
L       24.0.0.1/32 is directly connected, Ethernet0/3
S       32.0.0.0/8 [1/0] via 23.0.0.2
S       45.0.0.0/8 [1/0] via 24.0.0.2
S       172.36.0.0/16 [1/0] via 23.0.0.2
S       192.50.20.0/24 [1/0] via 12.0.0.1

```

R3

```
*Aug 26 11:47:41.255: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to down
*Aug 26 11:47:41.264: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to down
*Aug 26 11:47:42.259: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/2, changed state to down
*Aug 26 11:47:42.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/3, changed state to down
R3#
R3#
R3#
R3#con
R3#conf
R3#configure ter
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int e0/0
R3(config-if)#ip add 23.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#int e0/1
R3(config-if)#ip add 32.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#exit
R3#wr
*Aug 26 11:57:09.885: %SYS-5-CONFIG_I: Configured from console by console
R3#write
Building configuration...
[OK]
```

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 192.50.20.0 255.255.255.0 23.0.0.1
R3(config)#ip route 12.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 172.36.0.0 255.255.0.0 32.0.0.2
R3(config)#ip route 24.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 26.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 45.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 223.0.50.0 255.255.255.0 23.0.0.1
R3(config)#exit
R3#
```

Gateway of last resort is not set

```
S      12.0.0.0/8 [1/0] via 23.0.0.1
      23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      23.0.0.0/8 is directly connected, Ethernet0/0
L      23.0.0.2/32 is directly connected, Ethernet0/0
S      24.0.0.0/8 [1/0] via 23.0.0.1
S      26.0.0.0/8 [1/0] via 23.0.0.1
      32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      32.0.0.0/8 is directly connected, Ethernet0/1
L      32.0.0.1/32 is directly connected, Ethernet0/1
S      45.0.0.0/8 [1/0] via 23.0.0.1
S      172.36.0.0/16 [1/0] via 32.0.0.2
S      192.50.20.0/24 [1/0] via 23.0.0.1
S      223.0.50.0/24 [1/0] via 23.0.0.1
R3#
```

R4

```
R4#
R4#
R4#int e0/0
^
% Invalid input detected at '^' marker.

R4#con
R4#conf
R4#configure ter
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int e0/0
R4(config-if)#ip add 24.0.0.0 255.0.0.0
Bad mask /8 for address 24.0.0.0
R4(config-if)#no sh
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#int e0/1
R4(config-if)#ip add 45.0.0.1 255.0.0.0
R4(config-if)#no s
R4(config-if)#no sh
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#exit
R4#
*Aug 26 11:59:32.780: %SYS-5-CONFIG_I: Configured from console by console
R4#write
Building configuration...
[OK]
```

```
R4#configure te
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip route 172.36.0.0 255.255.0.0 45.0.0.2
R4(config)#ip route 192.50.200.0 255.255.255.0 24.0.0.1
R4(config)#ip route 223.0.50.0 255.255.255.0 24.0.0.1
R4(config)#ip route 12.0.0.0 255.0.0.0 24.0.0.1
R4(config)#ip route 23.0.0.0 255.0.0.0 24.0.0.1
R4(config)#ip route 32.0.0.0 255.0.0.0 45.0.0.2
R4(config)#ip route 26.0.0.0 255.0.0.0 24.0.0.1
R4(config)#exit
R4#wr
*Aug 26 12:29:56.034: %SYS-5-CONFIG_I: Configured from console by console
R4#wr
Building configuration...
[OK]
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S      32.0.0.0/8 [1/0] via 45.0.0.2
      45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      45.0.0.0/8 is directly connected, Ethernet0/1
L      45.0.0.1/32 is directly connected, Ethernet0/1
S      172.36.0.0/16 [1/0] via 45.0.0.2
R4#
```

R5

```
R5#
R5#
R5#
R5#con
R5#con
R5#con
R5#conf
R5#configure ter
R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#ip add 172.36.0.1 255.255.0.0
      ^
% Invalid input detected at '^' marker.

R5(config)#int e0/0
R5(config-if)#172.36.0.1 255.255.0.0
      ^
% Invalid input detected at '^' marker.

R5(config-if)#ip add 172.36.0.1 255.255.0.0
R5(config-if)#no sh
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#int e0/1
R5(config-if)#ip add 32.0.0.2 255.0.0.0
R5(config-if)#n sh
% Ambiguous command:  "n sh"
R5(config-if)#no sh
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#int e0/2
R5(config-if)#ip add 45.0.0.2 255.0.0.0
R5(config-if)#no sh
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
```

```
R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#ip route 23.0.0.0 255.0.0.0 32.0.0.1
R5(config)#ip route 24.0.0.0 255.0.0.0 45.0.0.1
R5(config)#ip route 26.0.0.0 255.0.0.0 45.0.0.1
R5(config)#ip route 12.0.0.0 255.0.0.0 32.0.0.1
R5(config)#ip route 192.50.20.0 255.255.255.0 32.0.0.1
R5(config)#ip route 223.0.50.0 255.255.255.0 45.0.0.1
R5(config)#exit
```

Gateway of last resort is not set

```
S    12.0.0.0/8 [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.2/32 is directly connected, Ethernet0/1
    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/2
L    45.0.0.2/32 is directly connected, Ethernet0/2
    172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.36.0.0/16 is directly connected, Ethernet0/0
L    172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
R5#
```

R6

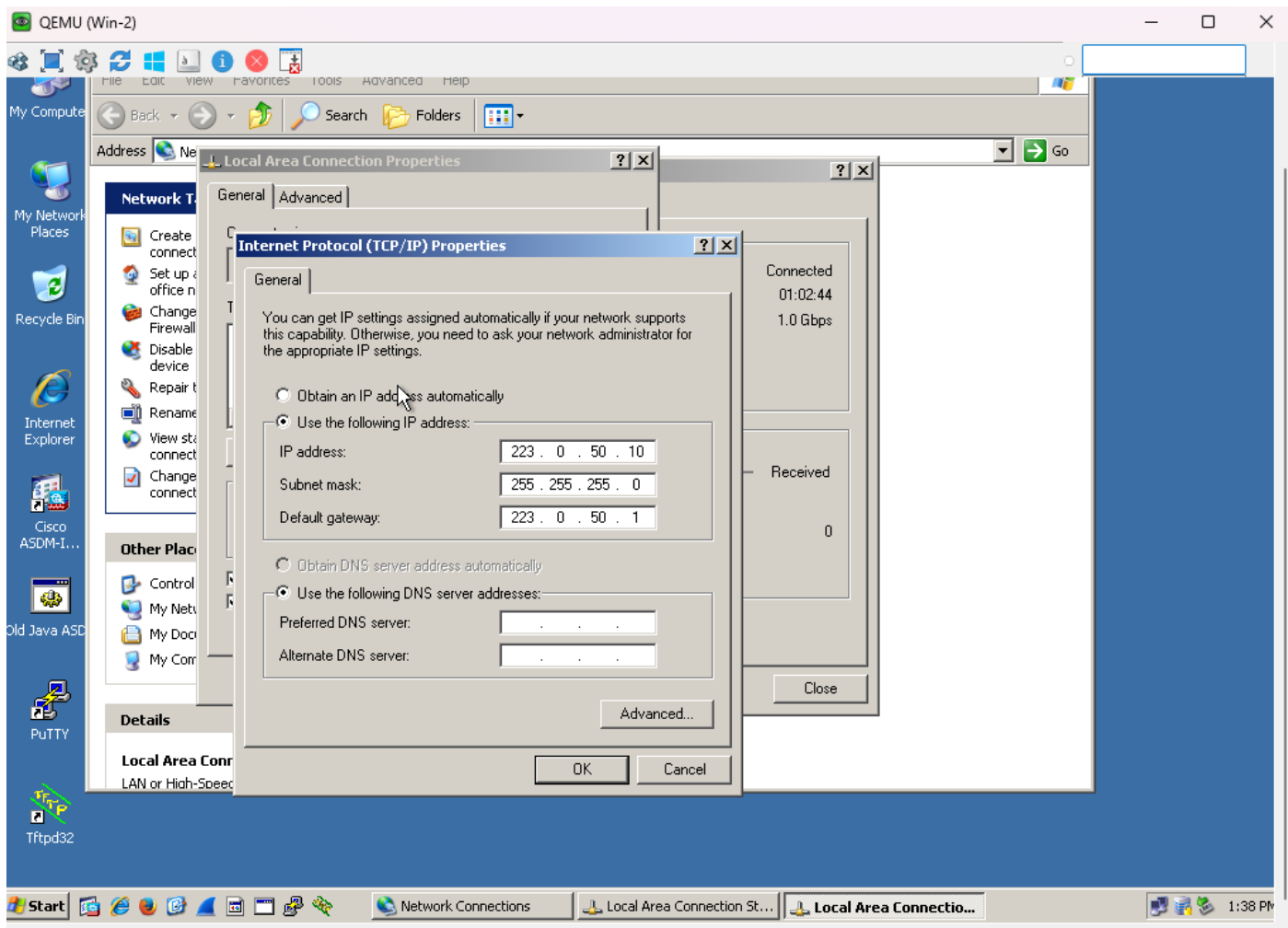
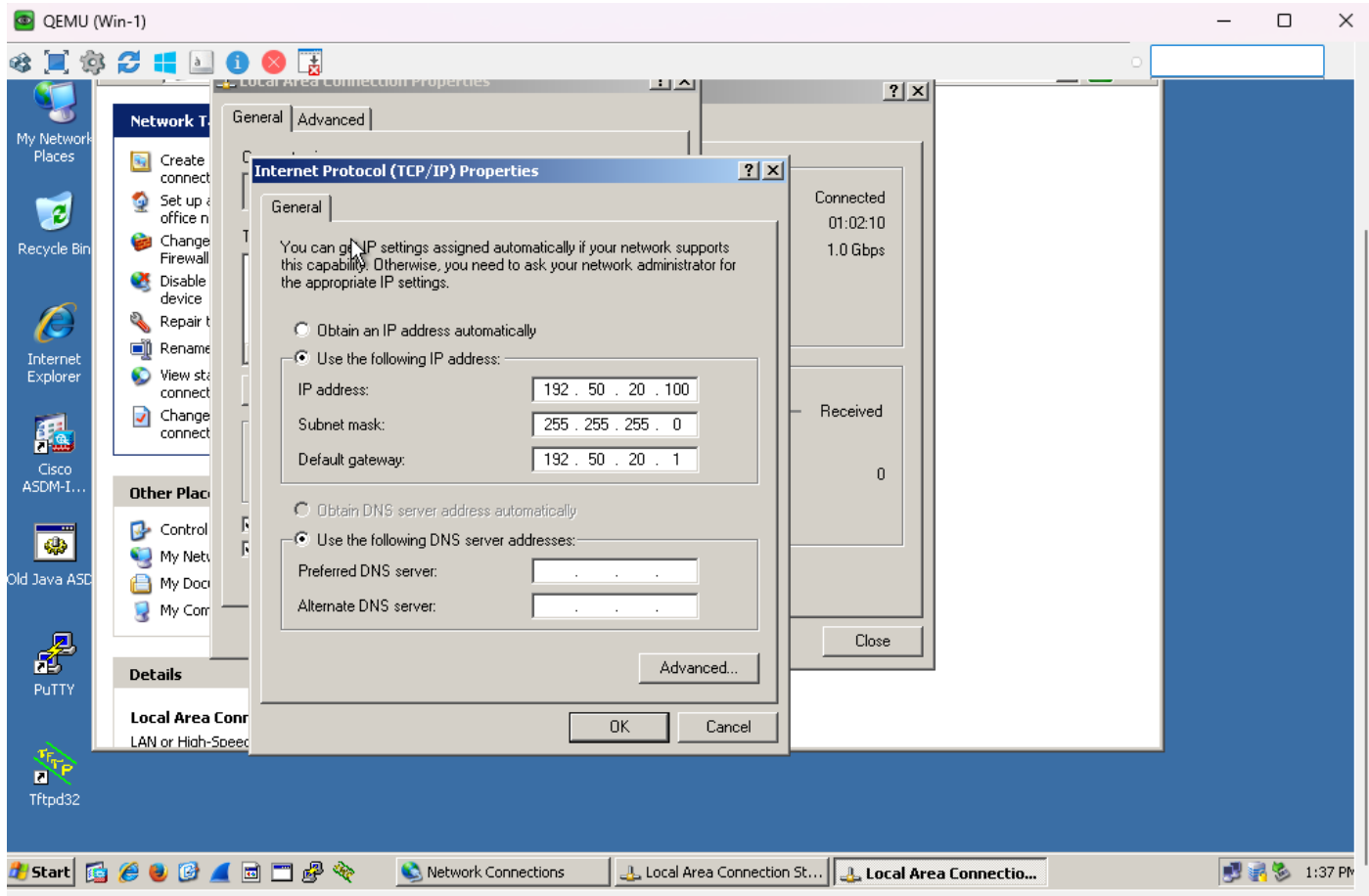
```
*Aug 26 12:03:54.023: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to down
*Aug 26 12:03:54.028: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
R6#
R6#
R6#
R6#
R6#conf
R6#configure ter
R6#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#int e0/0
R6(config-if)#ip add 223.0.50.1 255.255.255.0
R6(config-if)#no sh
R6(config-if)#no shutdown
R6(config-if)#
*Aug 26 12:08:42.674: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wiza
R6(config-if)#exit
R6(config)#int e0/1
R6(config-if)#ip add 26.0.0.2 255.0.0.0
R6(config-if)#no sh
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#exit
R6#wr
*Aug 26 12:09:24.678: %SYS-5-CONFIG_I: Configured from console by console
R6#write
Building configuration...
[OK]
R6#
```

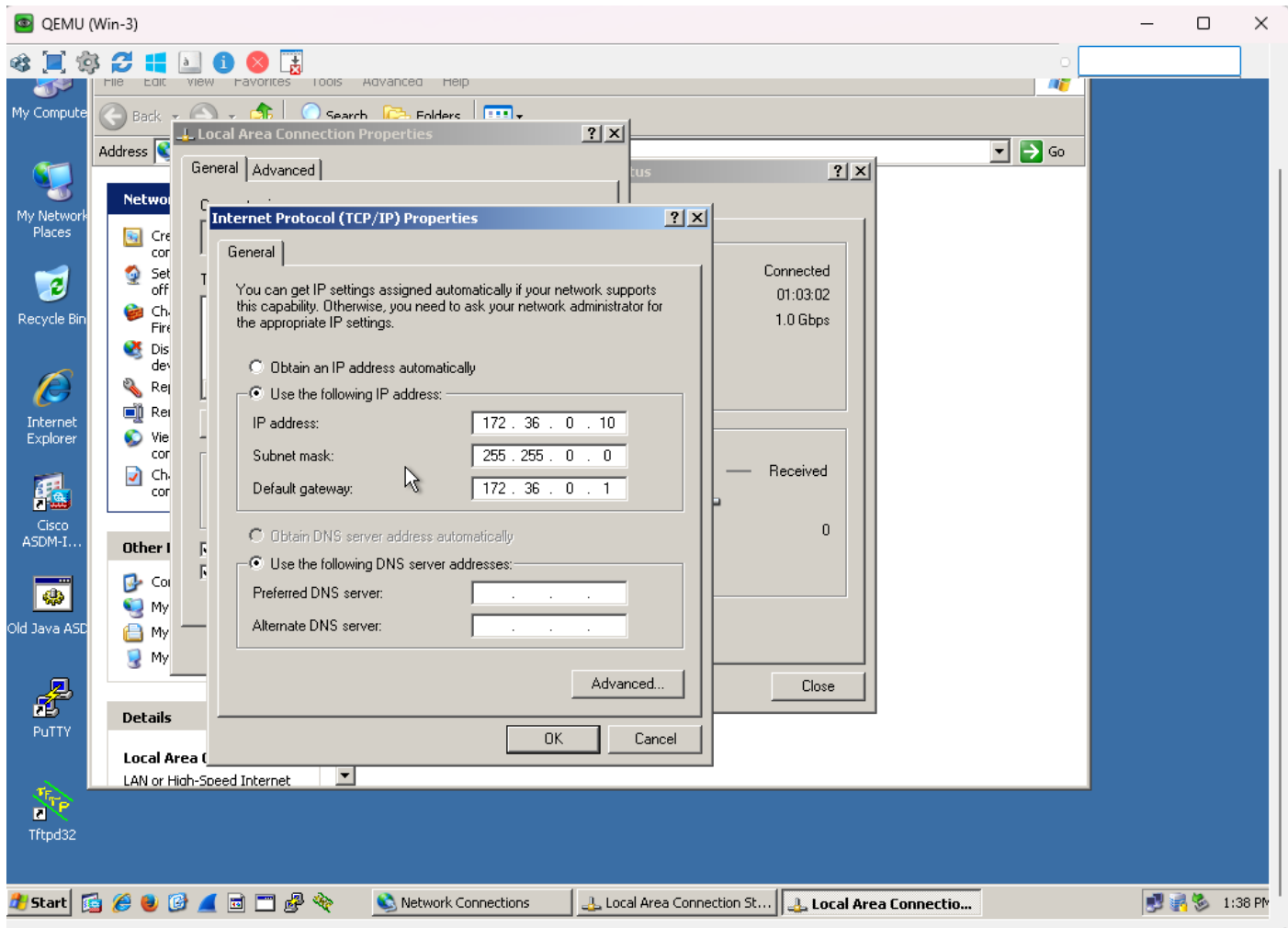
R6

```
R6(config)#ip route 192.50.20.0 255.255.255.0 26.0.0.1
R6(config)#ip route 172.36.0.0 255.255.0.0 26.0.0.1
R6(config)#ip route 12.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 23.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 24.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 32.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 45.0.0.0 255.0.0.0 26.0.0.1
R6(config)#exit
R6#w
*Aug 26 12:31:53.002: %SYS-5-CONFIG_I: Configured from console by console
R6#wr
Building configuration...
[OK]
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
S    26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      26.0.0.0/8 is directly connected, Ethernet0/1
L      26.0.0.2/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
S    223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C      223.0.50.0/24 is directly connected, Ethernet0/0
L      223.0.50.1/32 is directly connected, Ethernet0/0
```





```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=5ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 2ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=3ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=3ms TTL=125

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```


C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=1ms TTL=125

Reply from 192.50.20.100: bytes=32 time=2ms TTL=125

Reply from 192.50.20.100: bytes=32 time=2ms TTL=125

Reply from 192.50.20.100: bytes=32 time=2ms TTL=125

Ping statistics for 192.50.20.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>_

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=2ms TTL=124

Reply from 192.50.20.100: bytes=32 time=7ms TTL=124

Reply from 192.50.20.100: bytes=32 time=2ms TTL=124

Reply from 192.50.20.100: bytes=32 time=3ms TTL=124

Ping statistics for 192.50.20.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 7ms, Average = 3ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=3ms TTL=124

Reply from 223.0.50.10: bytes=32 time=2ms TTL=124

Reply from 223.0.50.10: bytes=32 time=2ms TTL=124

Reply from 223.0.50.10: bytes=32 time=2ms TTL=124

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_